ECON 0100 | Vignette C1 | Externalities

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

Hogsmeade has recently seen a surge in shops selling Glittering Gum, a candy that makes one's words sparkle and glisten as one speaks. The Demand curve and Supply curve for Glittering Gum can be represented using the following relationships:

$$D : P_b = 30 - \frac{1}{2}Q_d \qquad S : P_s = 6 + \frac{1}{4}Q_s$$

After years of careful epidemiological analysis, a subcommittee of the Ministry tasked with improving the health of the wizarding community published a story in the Daily Profit establishing a link between the consumption of Glittering Gum and "Starlight Sneeze," a condition where non-consumers start sneezing out harmless but bothersome glitter. The story estimated this condition carries an inconvenience cost of 6 Coins per unit to those in Hogsmeade.

Use a graph and algebra to find equilibrium price, quantity, and DWL.

Equilibrium Price ($\hat{P}$): _____ Equilibrium Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____ Deadweight Loss (DWL): _____

ECON 0100 | Vignette C2 | Taxes + Subsidies

Before the connection between Glittering Gum and Starlight Sneeze was established, the fiscal arm of the Ministry implemented a 3 Coin subsidy on every sale of Glittering Gum, to put a little sparkle into Hogsmeade's conversations. The Demand and Supply curves are the same as in Vignette C1.

With no known externalities, use a graph to illustrate the impact of the subsidy on the market.

Buyer's Price (P_b): _____ Seller's Price (P_s): _____

Subsidy Quantity (Q^{Sub}): _____ Dead Weight Loss (DWL): _____

ECON 0100 | Vignette C3 | Correctives

After the report on the connection between Glittering Gum and Starlight Sneeze, the Ministry of Magic has tasked it's economics branch to develop a policy to correct the problem.

Use a graph to propose a corrective policy and how it eliminates the DWL you found in Vignette C1.

Policy Type: _____ Policy Size: _____

Buyer's Price (P_b): _____ Seller's Price (P_s): _____

Post-Policy Quantity (Q^{τ}): _____ Dead Weight Loss (DWL): _____